

Supplementary Materials

BCHAIN-ACCESS: An Accessibility-by-Design Framework Integrating Blockchain and HCI for Healthcare Interfaces

This supplement provides implementation-depth and audit material referenced from the main paper. S1–S3 are development/feasibility artifacts recorded on a local Hardhat EVM node and are not representative of production-network behaviour. S4–S6 support the security, accessibility-conformance, and positioning claims. Cells marked **Confirm** indicate values the authors should finalise against their own audit logs and the public deposit (see the project repository).

S1. Development-Network Gas Benchmarks

Recorded on a local Hardhat EVM node (no congestion). ETH costs illustrative at 20 Gwei; *not* representative of mainnet, public-testnet, or Layer-2 costs.

Contract Function	EVM Gas	Est. ETH	Category
<code>registerRecord</code>	85,412	≈ 0.0017	Write (storage init)
<code>initiateGrant</code>	112,654	≈ 0.0022	Write (struct setup)
<code>confirmGrant</code>	47,210	≈ 0.0009	Modify (storage update)
<code>cancelPendingGrant</code> [†]	24,198	≈ 0.0005	Modify (pending cleared)
<code>revokeAccess</code>	42,504	≈ 0.0008	Modify (de-activation)
<code>checkAccess</code>	0	0 (free)	Read (view)
<code>getAuditLog</code>	0	0 (free)	Read (view)

[†]Backend-mediated reset of off-chain pending state, not an on-chain transaction; do not sum with on-chain rows. Only 32-byte hashes are stored on-chain.

S2. Threat-Vector-to-Mitigation Matrix

Threat vector	Implemented mitigation
Unauthorized access	<code>accessGrants</code> check + off-chain-encrypted records
Replay attacks	EVM account nonces
Coerced / accidental grants	60 s cooldown + <code>cancelPendingGrant()</code>
Audit-log tampering	Append-only <code>AuditEntry[]</code>
Sybil attacks	DAC whitelist model
Front-running of a pending grant	<code>onlyPatient(msg.sender)</code> guard on <code>confirmGrant</code>

S3. Feature-by-Feature Architecture Comparison

Property	Centralized EHR	Pure-Ethereum	BCHAIN-ACCESS
Patient data sovereignty	No	Partial	Yes
PII protection	DB-leak exposure	On-chain string exposure	On-chain hash only
Two-phase time-locked window	No	No	Yes
Immutable audit trail	No	Yes	Yes
Explicit WCAG-2.1-AA UI layer	No	No	Yes
Access-check latency	—	—	<10 ms (view , informal)

S4. Security Threat Model (STRIDE)

STRIDE decomposition of `HealthAccessControl.sol`. “Residual/next step” items are scoped to the independent audit (Limitation L8) and public-testnet evaluation (L7).

STRIDE category	Contract-level exposure	Mitigation / residual
Spoofing (identity)	Impersonating a patient or provider address	EVM account keys + <code>onlyPatient(msg.sender)</code> binding; residual: key-management/HSM out of scope
Tampering (data)	Altering records or grants	On-chain hash anchoring; append-only structures; off-chain records encrypted at rest (Confirm KMS)
Repudiation	Denying a grant/revoke occurred	Append-only <code>AuditEntry[]</code> audit log with immutable event trail
Information disclosure	Leaking PII via chain data	Only 32-byte hashes on-chain; personal data off-chain; salted digests
Denial of service	Blocking access or exhausting gas	<code>view</code> reads are free; bounded loops; residual: public-network gas/DoS under load unmeasured (L7)
Elevation of privilege	Gaining access without patient consent	Single-role <code>onlyPatient</code> modifier; two-phase time-locked confirmation; residual: static analysis (Slither/Mythril) recommended (see S6)

S5. WCAG 2.1 Level AA Conformance Coverage Matrix

Per-criterion coverage of the UI-layer prototype (Levels A + AA = 50 success criteria for AA conformance). Status: **Yes** = implemented with the stated mechanism; *N/A* = not applicable (no such content type); **Confirm** = present in the automated/code-review audit, authors to finalise against logs. (Media criteria 1.2.x are *N/A*: the prototype contains no audio or video.)

SC	Success Criterion (Level)	Status	Mechanism / note
1.1.1	Non-text Content (A)	Yes	Alt text on all images/icons
1.2.1–1.2.5	Time-based media (A/AA)	<i>N/A</i>	No audio/video content
1.3.1	Info & Relationships (A)	Yes	Semantic HTML + ARIA landmarks
1.3.2	Meaningful Sequence (A)	Yes	Logical DOM/reading order
1.3.3	Sensory Characteristics (A)	Yes	Instructions not colour/shape-only
1.3.4	Orientation (AA)	Confirm	Responsive layout
1.3.5	Identify Input Purpose (AA)	Confirm	Autocomplete tokens on inputs
1.4.1	Use of Colour (A)	Yes	Colour not sole indicator
1.4.2	Audio Control (A)	<i>N/A</i>	No auto audio
1.4.3	Contrast (Minimum) (AA)	Yes	29/29 pairs $\geq 4.5:1$ (3:1 large)
1.4.4	Resize Text (AA)	Confirm	Zoom to 200% without loss
1.4.5	Images of Text (AA)	Yes	Real text, not images of text
1.4.10	Reflow (AA)	Confirm	Reflow at 320 CSS px
1.4.11	Non-text Contrast (AA)	Yes	UI components/focus $\geq 3:1$
1.4.12	Text Spacing (AA)	Confirm	No clipping under spacing override
1.4.13	Content on Hover/Focus (AA)	Confirm	Dismissible/persistent tooltips
2.1.1	Keyboard (A)	Yes	Full keyboard operability
2.1.2	No Keyboard Trap (A)	Yes	Verified no trap
2.1.4	Character Key Shortcuts (A)	<i>N/A</i>	No single-key shortcuts
2.2.1	Timing Adjustable (A)	Confirm	60s cooldown — make user-configurable (open item; see main paper)
2.2.2	Pause, Stop, Hide (A)	<i>N/A</i>	No moving/auto-updating content

SC	Success Criterion (Level)	Status	Mechanism / note
2.3.1	Three Flashes (A)	Yes	No flashing; reduced-motion honoured
2.4.1	Bypass Blocks (A)	Yes	“Skip to main content” link
2.4.2	Page Titled (A)	Yes	Descriptive title
2.4.3	Focus Order (A)	Yes	Logical focus order
2.4.4	Link Purpose (In Context) (A)	Yes	Descriptive link text
2.4.5	Multiple Ways (AA)	Confirm	Nav + direct actions
2.4.6	Headings & Labels (AA)	Yes	Descriptive headings/labels
2.4.7	Focus Visible (AA)	Yes	Visible focus indicators
2.5.1–2.5.4	Pointer/Motion (A)	Confirm	No path/motion-only gestures
3.1.1	Language of Page (A)	Yes	lang attribute set
3.1.2	Language of Parts (AA)	N/A	Single-language content
3.2.1	On Focus (A)	Yes	No context change on focus
3.2.2	On Input (A)	Yes	No unexpected change on input
3.2.3	Consistent Navigation (AA)	Yes	Consistent nav across views
3.2.4	Consistent Identification (AA)	Yes	Consistent component labels
3.3.1	Error Identification (A)	Yes	Errors announced + identified
3.3.2	Labels or Instructions (A)	Yes	Labels/instructions on inputs
3.3.3	Error Suggestion (AA)	Confirm	Correction guidance on errors
3.3.4	Error Prevention (Legal/Fin/Data) (AA)	Yes	Two-stage confirmation + time-lock + cancel
4.1.1	Parsing (A)	Yes	Valid markup
4.1.2	Name, Role, Value (A)	Yes	ARIA name/role/value on components
4.1.3	Status Messages (AA)	Yes	ARIA live regions announce updates

Note. 2.2.1 (Timing Adjustable) is the one open conformance item: the 60s window is currently fixed and is flagged in the main paper as requiring user-configurability; two evaluators (E7, E9) independently raised it. It is the top remediation item for the next iteration.

S6. Static Analysis (Slither) — Results

A static analysis of `HealthAccessControl.sol` was performed using the Slither static analysis framework on 2026-07-08. The compiler used was Solidity v0.8.19 and Slither version was v0.11.5. No high- or medium-severity issues were found. The full analysis report is deposited in the public repository at `analysis/slither_report.json`.

Detector / severity	Count	Status	Resolution
High	0	Pass	No high-severity vulnerabilities found.
Medium	0	Pass	No medium-severity vulnerabilities found.
Low / Informational	52	Reviewed	Reviewed: 36 naming conventions, 10 block.timestamp comparisons for deliberate time-lock (acceptable), 2 string conversion helpers (false positives), 1 shadowing, 1 boolean comparison, 1 missing event, 1 solc version.
Optimization	0	Pass	None.